

emerged as a promising tool for operationalizing big data analytics and machine learning (ML) algorithms. Clinicians should be familiar with the basic concepts of ML as they apply to echocardiography.

BIG DATA

Big data are commonly characterized by the 4 Vs (**Figure 1A**): volume (digital amount of data); variety (diversity of sources that make up data); velocity (ability to detect data aspects, analyze them, and reach conclusion in real time); and veracity (differences in reliability among data elements). All of these exist in echocardiographic data (3). Each echocardiographic study consists of several acquisition modes, multiple views, and numerous frames generating large amounts of data. However, echocardiographic data commonly has noise, acquisition and measurement inconsistencies, as well as physiological variability, which raises concerns about potential errors with application of ML-based methods to heterogeneous echocardiographic databases. Rigorous quality assessment and control are essential before implementation of these methods in clinical practice. In contrast, other fields (e.g., genomics) have embraced ML, despite interfacing with noise, and have yielded important outcomes crucial to our understanding of disease processes and drug development. Standards and protocols have evolved to allow for high quality of such analyses (4). Therefore, ML algorithms can be arguably useful for extracting patterns while handling the noise within echocardiographic data.

LATENT AND HIDDEN VARIABLES

Latent variables or states cannot be observed directly but are rather inferred from other, directly measured variables. Certain variables can be “pseudo-latent” or “hidden,” meaning they are measurable, but have not been yet discovered. For example, echocardiographic strain variables were hidden until the advent of myocardial deformation imaging using color tissue Doppler (CTDI), which later evolved into speckle tracking echocardiography due to issues with noise and inaccuracies related to CTDI dependence on the insonation angle. It is unknown how many hidden variables exist within echocardiographic datasets, and the future of AI promises discovery of novel measurable variables. However, truly latent variables or states describe abstract concepts inferred from the measurable variables. In echocardiography, diastolic function is an important example of a latent state, which is indirectly inferred from measures of left ventricular relaxation, compliance, and filling pressures. The degree of refinement of the description of

diastolic function as a latent state is largely dependent on the number of hidden variables that can be continuously discovered and on the number of data points that can be accommodated by computational modeling (**Figure 1B**).

Methods for latent variable modeling do exist and include factor analysis, principal component analysis, and K-means clustering, among others.

In conclusion, echocardiographic data are suitable for ML algorithm-based research and applications. The appropriate use of AI tools promises reductions in cost, cognitive errors, and variability in clinical settings. These tools can also shed light on the complexity and nonlinear relationships in echocardiographic data beyond rigid traditional analytic approaches.

Alaa Mabrouk Salem Omar, MD, MSc, PhD

Chayakrit Krittanawong, MD

Sukrit Narula, MD

Jagat Narula, MD, PhD

Edgar Argulian, MD, MPH*

*Mount Sinai St. Luke's Hospital

Mount Sinai Heart

Icahn School of Medicine at Mount Sinai

1111 Amsterdam Avenue

New York, New York 10025

E-mail: Edgar.Argulian@mountsinai.org

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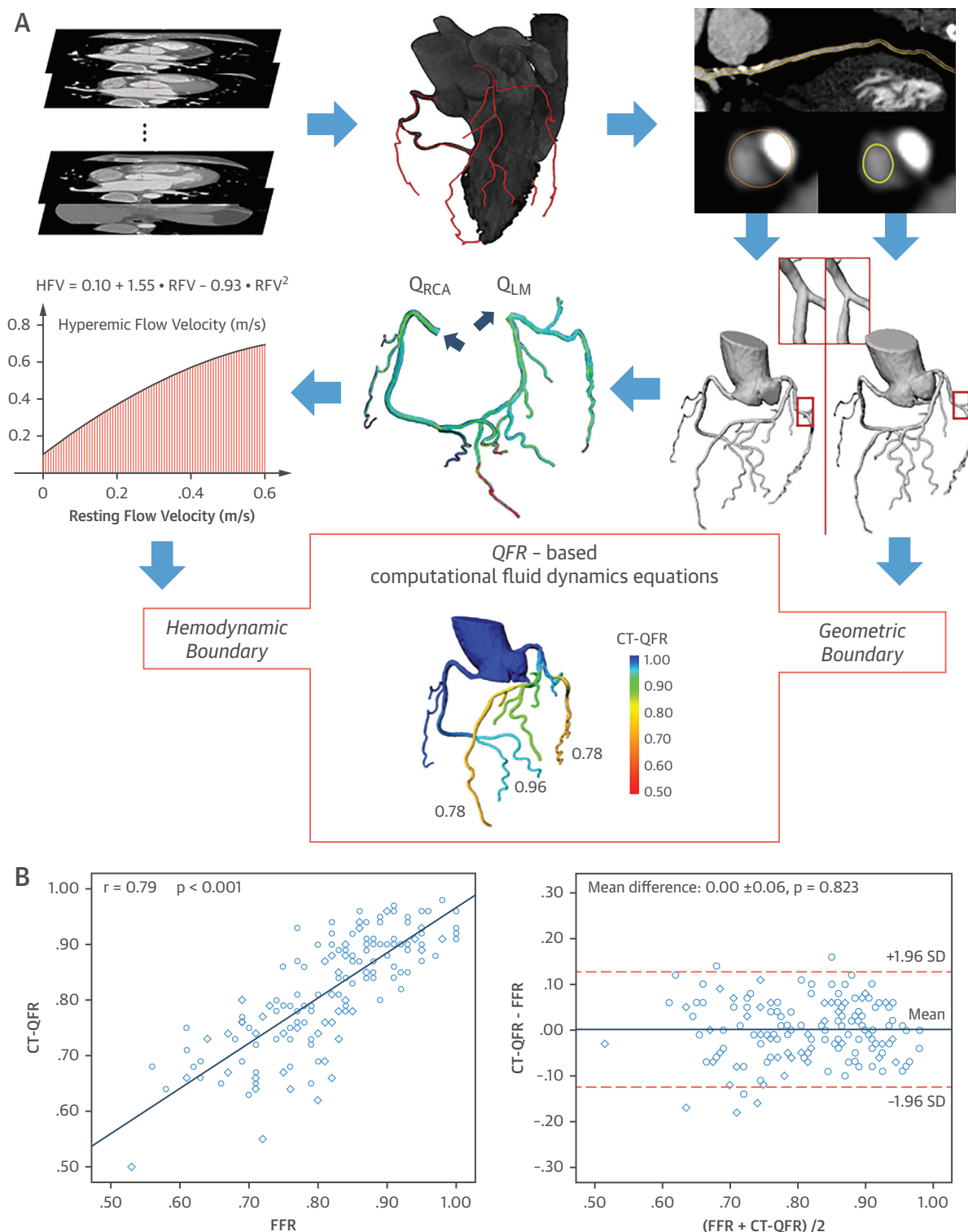
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Diagnostic Accuracy of a Fast Computational Approach to Derive Fractional Flow Reserve From Coronary CT Angiography



Fractional flow reserve (FFR) is the gold standard to assess the physiological significance of coronary stenoses. Quantitative flow ratio (QFR) is a novel algorithm to compute FFR from imaging data (1). Good diagnostic concordance between QFR and FFR was

FIGURE 1 Computation of CT-QFR and Comparison With FFR



(A) Framework of computed tomography angiography–based quantitative flow ratio (CT-QFR) computation and **(B)** the correlation and agreement between CT-QFR and fractional flow reserve (FFR).

observed when applied to invasive coronary angiography (2). However, the accuracy of QFR when applied to noninvasive coronary computed tomography angiography (CTA) is unknown. The objective of this study was to evaluate the feasibility and diagnostic accuracy of CTA-based QFR (CT-QFR), using wire-based FFR as the reference standard.

Figure 1A shows the framework of CT-QFR computation. Coronary arterial lumens were delineated in a semi-automatic manner (3) by an experienced analyst blinded to invasive FFR values. All coronary segments with reference lumen diameter of ≥ 1.5 mm were analyzed and subsequently exported into a prototype software package (Cta-Plus, Pulse Medical Imaging Technology, Shanghai, China), in which coronary tree reconstruction and CT-QFR computation were automatically performed. First, all coronary arteries were merged into a hierarchical tree structure, and the reference lumen (i.e., the healthy lumen), as if there was no stenosis, was reconstructed. Subsequently, patient-specific resting coronary flow was estimated based on its three-fourths powered function, with corresponding normal arterial volume derived from the allometric scaling law, followed by conversion to virtual hyperemic flow (1). The reference lumen, instead of the actual lumen geometry, was used to estimate the flow, because the maximally demanded flow by the subtended myocardial mass would not change with the developed epicardial stenosis. Finally, the virtual hyperemic flow was used to compute CT-QFR value at every position of the reconstructed coronary tree, using the previously validated QFR algorithm (2).

FFR was measured using a 0.014-inch pressure guidewire under the hyperemic condition induced by intravenous administration of adenosine (140 $\mu\text{g/kg/min}$) via the antecubital vein. The CT-QFR value at the end of the interrogated vessel was used to compare with FFR.

This retrospective and observational study included 154 consecutive patients with stenoses found in the major epicardial coronary arteries by CTA, who were also referred to clinically indicated invasive coronary angiography and FFR measurements within 1 month at Shanghai Jiao Tong University Affiliated Sixth People's Hospital, Shanghai, and Beijing Anzhen Hospital, Capital Medical University, Beijing. Coronary CTA images were acquired using ≥ 64 slice CT scanners. Patients with poor image quality, previous percutaneous coronary intervention or coronary artery bypass grafting on the interrogated vessel and presence of collateral circulation were excluded (13.0%).

A total of 156 vessels (64.1% left anterior descending arteries) from 134 patients were finally included for CT-QFR analysis. The interrogated vessels had an average FFR of 0.82 ± 0.10 , with 62 (39.7%) vessels in the range between 0.75 and 0.85. Ninety (57.7%) vessels had bifurcation lesions. Good correlation ($r = 0.79$; $p < 0.001$) and agreement (0.00 ± 0.06 ; $p = 0.823$) between CT-QFR and FFR was observed (**Figure 1B**). Using ≤ 0.80 to define ischemia for both CT-QFR and FFR, vessel-level diagnostic accuracy, sensitivity, specificity, positive predictive value, and negative predictive value were 87.2%, 87.7%, 86.8%, 82.6%, and 90.8%, respectively. Patient-level diagnostic metrics were similar, being 87.3%, 90.2%, 84.9%, 83.3%, and 91.2%, respectively. CT-QFR significantly outperformed quantitative CTA-derived diameter stenosis (AUC: 0.92 vs. 0.76; $p < 0.001$) in diagnosis of ischemia.

The specificity and positive predictive value of CT-QFR appeared to be notably higher compared with previous CT-FFR studies, which indicated that less false positive cases would be encountered when using this novel method. In patients with FFR < 0.75 or > 0.85 ($n = 94$; 60.3%), CT-QFR demonstrated excellent diagnostic performance, which indicated a high confidence to either identify (sensitivity: 100.0%) or exclude (specificity: 98.3%) physiological significant lesions.

The presence of extensively calcified lesions ($n = 42$; 26.9%) defined by the combination of a cross-sectional calcium arc $> 90^\circ$ and a thickness > 1.5 mm influenced the diagnostic accuracy (81.0% vs. 89.5%; $p = 0.007$) and analysis variability (0.01 ± 0.06 vs. 0.01 ± 0.04 ; $F = 2.20$; $p = 0.013$). CT-QFR might differ more from FFR when severe tandem lesions are present. The distance between the 2 consecutive lesions was not long enough for the flow to be fully developed. Despite this, the present study indicated high reproducibility and diagnostic performance of CT-QFR in a clinical population. Notably, delineation of lumen borders for the entire coronary tree took 17.1 ± 7.6 min per patient. However, the subsequent CT-QFR computation took only 18.9 ± 8.5 s on an off-the-shelf computer without requiring large computational power, facilitating its implementation in routine functional evaluation of coronary stenosis during CTA imaging.

Zehang Li, BSc
Jiayin Zhang, MD†
Lei Xu, MD
Wenjie Yang, MD
Guanyu Li, BSc
Daixin Ding, BSc

Yunxiao Chang, MSc
Mengmeng Yu, MD
Pieter Kitslaar, MSc
Su Zhang, PhD
Johan H.C. Reiber, PhD
Armin Arbab-Zadeh, MD
Fuhua Yan, MD
Shengxian Tu, PhD*

*Med-X Research Institute
Room 123
Shanghai Jiao Tong University
No. 1954, Huashan Road
Shanghai 200030
PR China
E-mail: sxtu@sjtu.edu.cn
OR

†No. 600, Yishan Road
Shanghai 200233
PR China or andrewssmu@msn.com
<https://doi.org/10.1016/j.jcmg.2019.08.003>

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Prevalence and Prognostic Impact of Ascending Aortic Dilatation in Patients Undergoing TAVR



Ascending aortic dilatation occurs frequently in patients with aortic stenosis (AS) (1). In our study, we sought to investigate the prevalence, characteristics, and clinical impact of ascending aortic dilatation in patients undergoing transcatheter aortic valve replacement (TAVR) for tricuspid AS using detailed computed tomography (CT) data.

A total of 1,426 patients with tricuspid AS who underwent TAVR between January 2013 and November 2017 were included. All patients were evaluated using electrocardiogram-gated, contrast-

enhanced CT before TAVR using a second-generation dual-source CT system (Siemens SOMATOM Definition Flash; Siemens Healthcare, Erlangen, Germany). CT images (mid-systole) were reconstructed using 3mensio Valves software version 9.0 (3mensio Medical Imaging BV, Bilthoven, the Netherlands) and analyzed by a dedicated core laboratory at our institution. The maximal cross-sectional diameters of the ascending aorta at the level of the sinus of Valsalva, sinotubular junction, and tubular ascending aorta (from the sinotubular junction to the tubular tract at the level of the pulmonary artery bifurcation) were measured on a plane strictly perpendicular to the main axis of the aorta. Aortic root angulation was calculated from a coronal projection at the level of the aortic annulus and was defined as the angle between the horizontal plane and the plane of the aortic annulus (2). A threshold of 850 Hounsfield units was used to detect the leaflet calcium volume, as previously validated for the prediction of paravalvular regurgitation (3). Dilatation of the ascending aorta was defined as a maximal ascending aortic diameter of ≥ 40 mm, in accordance with previous studies (1,4). Follow-up data were recorded during outpatient visits or telephone interview at 30 days, 6 months, 12 months, and yearly thereafter. The Kaplan-Meier method was used to analyze unadjusted survival, which was compared using the log-rank test. The effect of ascending aortic dilatation on mortality was assessed using Cox proportional hazards models. Clinically relevant variables with a p value < 0.10 according to the univariable analysis were included in the multivariable model. The association between the maximal ascending aortic diameter and 2-year all-cause mortality was assessed using restricted cubic spline analysis.

In the overall cohort, the ascending aortic diameter was ≥ 40 mm in 13.7% of patients. Patients with a dilated ascending aorta were more likely to be men and have previous stroke or transient ischemic attack, atrial fibrillation, larger body surface area, and larger body mass index. Patients with a dilated ascending aorta had a lower left ventricular ejection fraction ($54.4 \pm 15.3\%$ vs. $57.8 \pm 14.5\%$; $p = 0.003$), a lower aortic valve mean gradient (41.3 ± 13.8 mm Hg vs. 43.5 ± 13.5 mm Hg; $p = 0.04$), and a higher left ventricular mass index (124.2 ± 40.2 g/m² vs. 111.1 ± 34.1 g/m²; $p < 0.001$). CT measurements showed that the leaflet calcium volume (850 Hounsfield units) and aortic root angulation were significantly higher in the dilated ascending aorta group (leaflet calcium volume: median 250.8 [interquartile range: 129.3 to 408.5] mm³ vs. 141.3 [interquartile range: 64.1 to 266.8] mm³; $p < 0.001$; aortic root